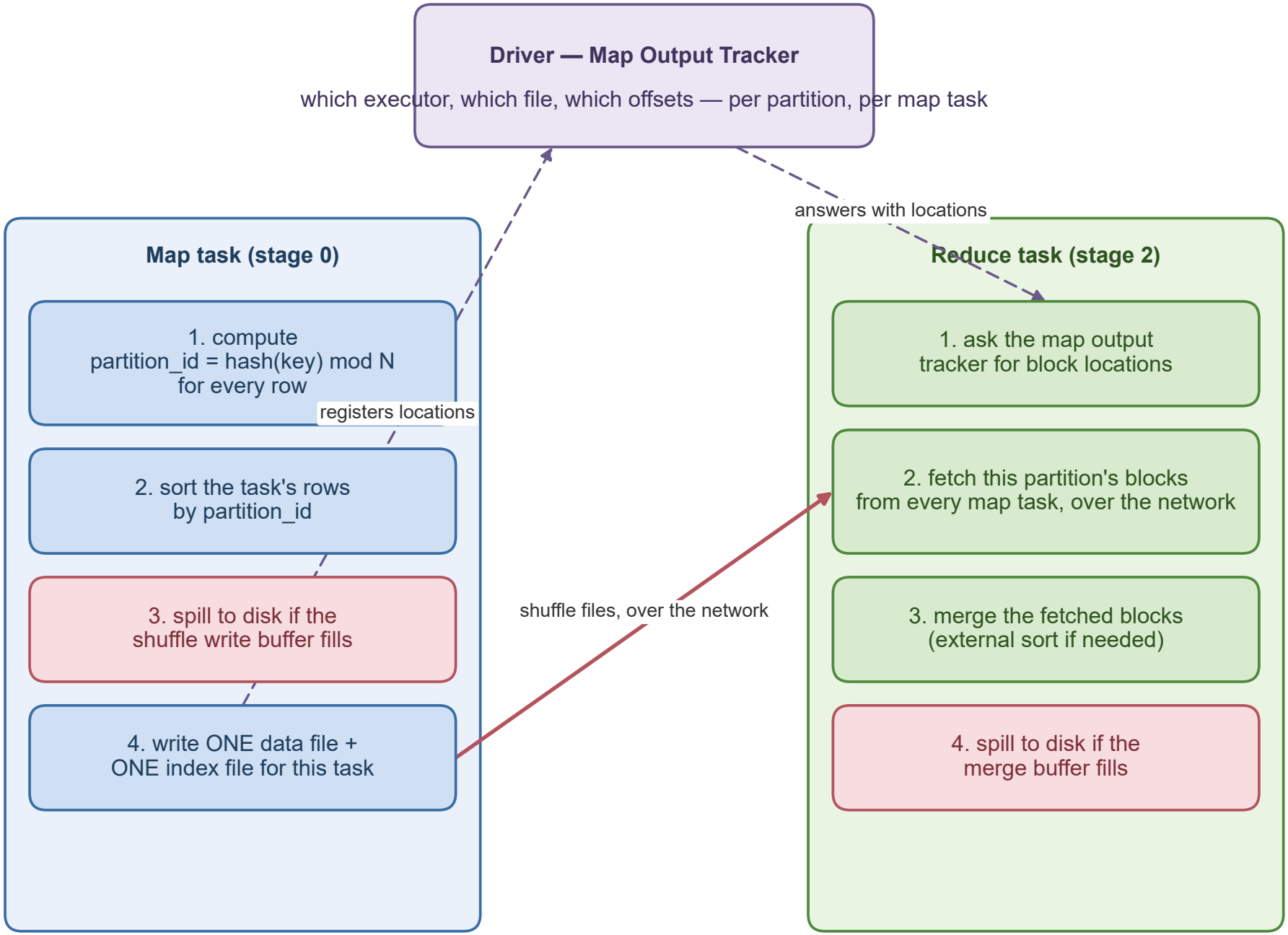

Driver — Map Output Tracker
which executor, which file, which offsets — per partition, per map task
answers with locations
Map task (stage 0)
1. compute
partition_id = hash(key) mod N
for every row
registers locations
2. sort the task's rows
by partition_id
3. spill to disk if the
shuffle write buffer fills
4. write ONE data file +
ONE index file for this task
shuffle files, over the network
Reduce task (stage 2)
1. ask the map output
tracker for block locations
2. fetch this partition's blocks
from every map task, over the network
3. merge the fetched blocks
(external sort if needed)
4. spill to disk if the
merge buffer fills